

Optimal pressure management in water networks: increased efficiency and reduced energy costs

Matteo Nicolini

Department of Chemistry, Physics and Environment

University of Udine

Udine, Italy

Email: matteo.nicolini@uniud.it

Abstract—In this paper we address the problem of pressure management in water distribution systems. We present a methodology based on single- and multi-objective genetic algorithms (GAs) which, starting from the numerical model of the network, performs calibration and then optimizes the location and control of pressure reducing valves (PRVs), according to some operational constraints that must be satisfied. In particular, the simulation model is calibrated with a real-coded, single-objective GA in order to obtain, on one hand, pipe roughness coefficient values and, on the other, an estimate of the subdivision of the total flow supplied to the network between actual customers demand and water loss. The multi-objective NSGA-II is implemented in order to find the Pareto optimal solutions representing different level of compromise between installation costs and leakage reduction. The application of the methodology to a real network allowed considerable water savings, as evidenced by the monitoring of the system. Two more advantages are also evident: firstly, the number of interventions for repairing pipes is more than halved, due to the reduced pressure regime (thus enabling the water utility to provide better service more efficiently and reliably); secondly, the surplus water is being diverted to a storage tank of a pumped network, thus allowing a notable reduction of pumping costs.

Keywords—Water supply, leakages, optimization, valves.

I. INTRODUCTION

Pressure control in water distribution networks is a major concern for water utilities all around the world. On one hand, systems characterized by high pressures suffer from background leakages (which tend to increase with time) and may be subjected to intolerable pipe break rates (with related service interruptions and customers complaints); on the other, low pressure regimes often do not guarantee an adequate service to customers, and are often a consequence of the increasing level of leakage in the system.

Keeping an adequate pressure throughout the network is a difficult task, especially in rapidly growing cities or in situations with remarkable differences in ground elevations. In addition, a key point to be considered is that water distribution systems are always designed in order to provide some standard pressures even under peak demand (or fire flow) conditions, thus constraining the network under excessive pressures during their lifetime.

As a result, energetic costs to manage a water distribution network tend to increase with time, very often leading to unsustainable situations that have to be faced. Water utilities

can reduce electricity use by increasing pumping efficiency, for example by replacing inefficient motors on pumps, or by installing variable-speed drives and implementing operational controls such as those provided through SCADA (Supervisory Control And Data Acquisition) systems.

Anyway, pumping less water is even more effective in reducing electricity. In order to reduce leakages, pressure management is now recognized as one of the most efficient and cost effective measures available to water utilities [1], [2], [3], [4]. Replacing ageing infrastructures also reduces water loss and decreases water flow resistance due to corrosion and mineral build-up on pipe walls, thus lowering energy for pumping, but the investments required are often one order of magnitude higher.

In order to rehabilitate a system, water utilities have to make decisions, often under a high degree of uncertainty. To help decision makers, hydraulic simulation models are widely adopted for analyzing the behavior of the network under different working conditions, but the predictive ability of a numerical model strongly depends on its calibration.

From a mathematical point of view, both calibration and optimal pressure management may be considered as optimization problems with non-linear objective functions and constraints.

In this paper, we consider a single-objective, real coded GA for solving the calibration problem [5], while we formulate pressure management as a two-objective optimization problem, in which the first criterion is the total number of pressure reducing valves (a surrogate for installation costs), while the second is represented by the total leakage in the system. In particular, we implemented the multi-objective NSGA-II (Non-dominated Sorting Genetic Algorithm, [6]), in order to optimize the two conflicting criteria. In addition, the particular coding of the real variables allows the determination of both the location and the regulation of the valves, according to a number of predefined demand conditions [7].

The application of the methodology to a real network resulted in considerable water and energy savings, due to the fact that the surplus water arising from leakage reduction is being diverted to the storage tank of a pumped system, thus reducing costs for pumping.

The paper is organized as follows: section II describes the methodology, section III presents the application to a real network, while section IV draws some concluding remarks.

II. METHODOLOGY

A. Model calibration

Model calibration is based on a real-coded GA in which the objective function is defined as the minimization of the weighted sum of maximum absolute differences between observed and calculated values:

$$\min : f = W_h \cdot \max_{n,t} |H_n^{obs}(t) - H_n^{calc}(t)| + W_p \cdot \max_{p,t} |Q_p^{obs}(t) - Q_p^{calc}(t)| \quad (1)$$

where $H_n^{obs}(t)$ and $H_n^{calc}(t)$ represent, respectively, the observed and calculated head at time t in node n , while $Q_p^{obs}(t)$ and $Q_p^{calc}(t)$ the observed and calculated flow at time t in pipe p . W_h and W_p are weighting factors for heads and flows, respectively.

The optimization problem is subjected to constraints defined by continuity equations ($\forall$ node i):

$$\sum_j Q_{ij}(t) - \alpha(t)Q_{b,i} - \ell_i(t) = 0, \quad \ell_i(t) = c_i p_i(t)^\gamma \quad (2)$$

and energy loss equations ($\forall$ link ij):

$$H_i(t) - H_j(t) = h_{ij}(t), \quad h_{ij}(t) = \frac{10.668 Q_{ij}(t)^{1.852} L_{ij}}{C_{ij}^{1.852} D_{ij}^{4.871}} \quad (3)$$

where $Q_{ij}(t)$ indicates the flow from node i to node j at time t , $Q_{b,i}$ the (mean) metered consumption at node i (which can be calculated by billing information), $\alpha(t)$ a multiplier for nodal demands depending on time, $\ell_i(t)$ the leakage at node i , $p_i(t)$ the pressure at node i , while c_i and γ are two coefficients quantifying the relationship between water loss and pressure at node i . In particular, we assumed a unique typology of leakage, that is, the same value of γ for all nodes, and we express the coefficient c_i as:

$$c_i = c \frac{Q_{b,i}}{Q_b} = \frac{Q_{b,i}}{\sum_i Q_{b,i}} \quad (4)$$

In (3), $H_i(t)$ and $H_j(t)$ are the total heads at nodes i and j at time t , while $h_{ij}(t)$ is the head loss between nodes i and j ; L_{ij} , D_{ij} and C_{ij} are the length, diameter and Hazen-Williams friction factor for pipe connecting nodes i and j , respectively. We chose the real coding of the decision variables, representing the Hazen-Williams pipe friction factors (four in this case) and the coefficient c , Fig. 1. Hydraulic constraints are guaranteed by Epanet 2 [8], to which the optimization model has been coupled.

B. Optimal pressure management

The problem of optimal pressure management in a water distribution system is addressed through the introduction and regulation of pressure reducing valves. The determination of the number of valves, together with their location and setting, is formulated as a two-criterion optimization problem, and is based on the multi-objective genetic algorithm previously developed [7].

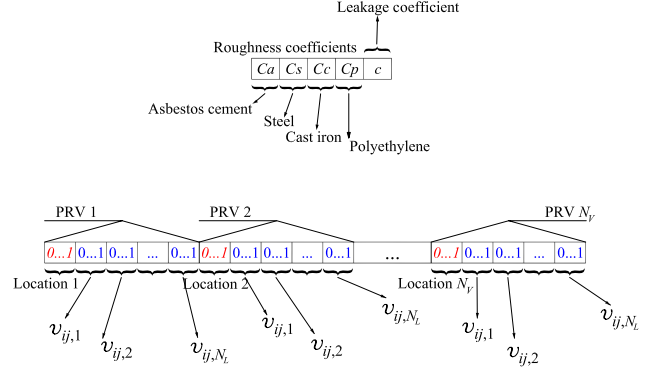


Fig. 1. String coding in the genetic algorithms implemented: real coding of pipe roughness coefficients and leakage coefficient adopted in model calibration (above); real coding for NSGA-II for multi-objective pressure management (below). The notation $0 \dots 1$ indicates a real number in the range between 0 and 1. The meaning of the symbols is the following: $v_{ij,k}$, diameter multiplier simulating the presence of a valve in link connecting nodes i and j for load condition k ; N_L , number of load (demand) conditions; N_V , maximum number of valves allowed.

The problem may be mathematically formulated as:

$$\min : f_1 = n_V \quad (5)$$

$$\min : f_2 = \frac{1}{N_L} \sum_{k=1}^{N_L} \sum_{i=1}^{N_S} w_k c_i p_{i,k}^\gamma \quad (6)$$

where n_V is the number of valves in a generic solution, N_S the number of nodes in the system and N_L the number of load (demand) conditions, each characterized by a weight, w_k .

Constraints are represented by three sets of equations: continuity for each node, hydraulic head loss relationship for each link (pipe or valve), and operational constraints, that is, the requirement that at each node pressure is for every load condition above a given level and the limit on the maximum number of allowed valves, N_V . With the same meaning of the symbols, but now with reference to load condition k instead of variation in time, t , we have:

- 1) Continuity at every node (conservation of mass), expressed as:

$$\sum_j Q_{ij,k} - \alpha_k Q_{b,i} - \ell_{i,k} = 0 \quad \text{for } k = 1, \dots, N_L \quad (7)$$

where the sum is extended to all links connected to node i . The subscript k refers to the k -th load (demand) condition.

- 2) Headloss in every link (conservation of energy), expressed as:

$$H_{i,k} - H_{j,k} = h_{ij,k} \quad (8)$$

According to the type of link, several expressions are possible for the head loss $h_{ij,k}$; in particular, we adopted the following equations useful to describe both pipes and pressure reducing valves.

- Pipes (Hazen-Williams formula in SI units):

$$h_{ij,k} = 10.668 C_{ij}^{-1.852} d_{ij}^{-4.871} L_{ij} Q_{ij,k}^{1.852} \quad (9)$$

- Pressure reducing valves:

$$h_{ij,k} = 10.668 C_{ij}^{-1.852} (v_{ij,k} d_{ij})^{-4.871} L_{ij} Q_{ij,k}^{1.852} \quad (10)$$

where $v_{ij,k}$ is a diameter multiplier (in the range $0, \dots, 1$) which simulates the presence of a valve in the link connecting nodes i and j , and is dependent on load condition k .

3) Operational constraints:

- Required pressure at each node:

$$p_{i,k} \geq p_{req,i} \quad (11)$$

- Maximum number of valves:

$$n_V \leq N_V \quad (12)$$

where N_V represents a pre-defined chosen value.

III. APPLICATION TO A REAL SYSTEM

The methodology has been applied to the water distribution system of Buja (in the north-east part of Italy, Fig. 2). The whole system is composed of two networks (high and low), separated by closed gate valves: the high network is supplied by a main line and a gravity storage tank, while the low network before the application of pressure management was entirely served by a pumped storage tank. Due to the morphology of the area, the high network was also the one characterized by the highest values of pressure (the average value was 7.5 bar, and in some points above 10 bar).

The high distribution network serves about 3500 inhabitants and is composed of several pipe materials: 45% of conduits is made of asbestos cement, 36% of steel, 10% of cast iron and 9% of polyethylene, for a total length of pipes of 35500 m.

Objective of the work was to reduce water loss in the high network through pressure management, and to transfer excess water coming from the main supply line to the low storage tank, in order to reduce electricity costs.

The simulation model was calibrated considering five decision variables [5]: four Hazen-Williams friction factors and the coefficient c , introduced in (4).

Once the model had been calibrated, we applied the pressure management algorithm with a maximum number of allowed valves, N_V , equal to 10, and two load conditions, $N_L = 2$, representing respectively the minimum and maximum user demand (according to seasonal and diurnal variations).

At first we fixed at 2 bar the required level of pressure at every node in the system, thus resulting in an actual average pressure surplus of 5.5 bar.

In all the runs the NSGA-II algorithm used a population of 100 individuals evolving for 1000 generations. Uniform

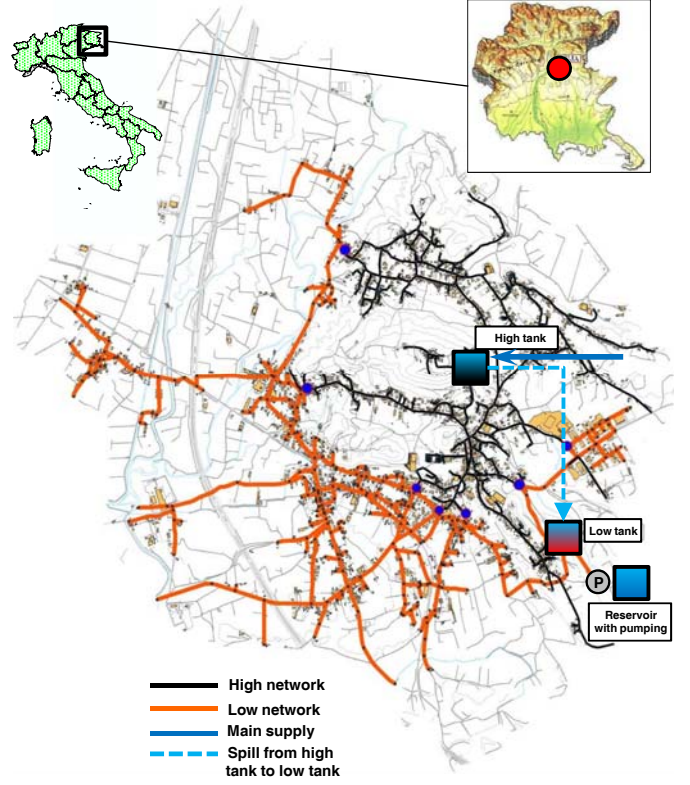


Fig. 2. General view of Buja water distribution system, with distinction between the high (in black) and the low network (in red).

crossover probability was equal to 0.8, while mutation probability was equal to the inverse of the string length (Fig. 1):

$$p_m = \frac{1}{(N_L + 1)N_V} = \frac{1}{30} \quad (13)$$

in which p_m is the probability of mutation at the gene level.

The water utility managing the system (CAFC S.p.A.) adopted one of the optimal four-valve solutions identified by the algorithm, and in the period July-September 2008 four PRVs were installed in the network, as shown in Fig. 3. In particular, pipes (265,410,577,663) were chosen to place the PRVs. The monitoring of the reservoir in these two years after the installation of the valves (Fig. 4) allowed to give a precise estimate of the value of the water savings: from Table I it is evident the effect of the PRVs in reducing water supplied to the high network. In particular, the increment in water and energy savings during the year 2010 is due to a fine-tuning of PRV settings, which have been regulated in order to guarantee a minimum pressure between 1.5 and 1.8 bar in some portions of the network (instead of 2.0 bar initially fixed).

The surplus water was transferred to the low network, thus achieving nearly 38% of energy saving in the first year (2009) and 54% during the second year (2010), due to the reduced pumped volumes.

Another main advantage arising from pressure management is the reduction of the number of interventions for pipe failures

TABLE I

WATER SUPPLIED TO THE HIGH NETWORK, ENERGY FOR PUMPING IN THE LOW NETWORK, AND NUMBER OF INTERVENTIONS FOR PIPE FAILURE OR JOINT BREAKS IN THE HIGH NETWORK.

Year	2008	2009	2010
Average flow supplied (L/s)	18.93	13.99	10.47
Volume supplied ($\text{m}^3 \cdot 10^3$)	597.13	441.18	330.18
Energy for pumping (MWh)	130.34	82.71	60.91
Interventions on high network	95	67	45

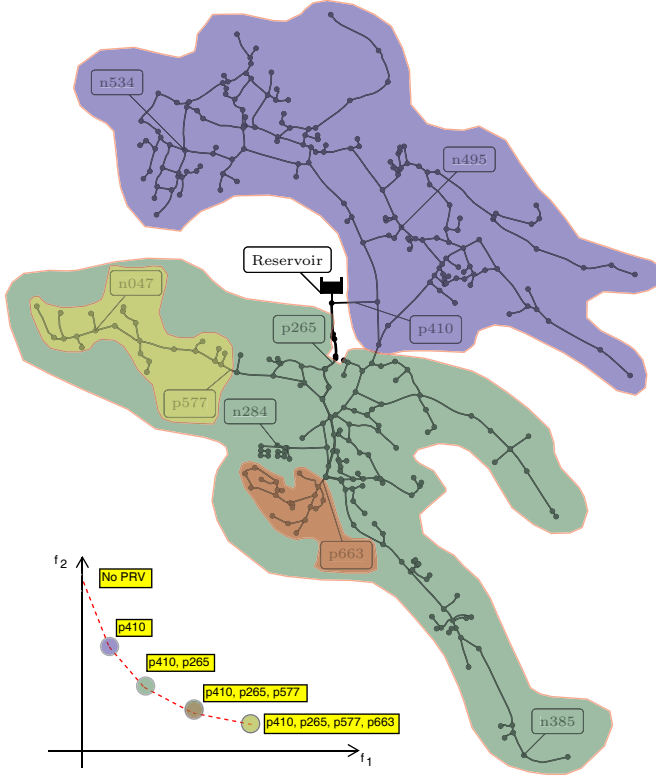


Fig. 3. High network model: location of manometers used for calibration (prefix *n*), and pipes selected for PRV installations (prefix *p*). Areas under the influence of each PRV are also indicated. The lower left corner shows a schematic representation of the Pareto front with the optimal solutions representing different tradeoff between the two objective functions.

or joint breaks, as evidenced in Table I: such opportunity allows the water utility to manage the system in a much more efficient way, thus avoiding numerous service interruptions which often lead to undesirable customers complaints.

IV. CONCLUSION

In this paper we focused on optimal pressure management in water distribution networks, which is now recognized as one of the most efficient and cost effective measures for reducing real losses and operational costs. We presented a methodology based on genetic algorithms, in particular using a single-objective GA for model calibration and NSGA-II in order to solve the multi-objective problem characterized by the following two conflicting criteria: the minimization of the number of PRVs and the minimization of the total water loss

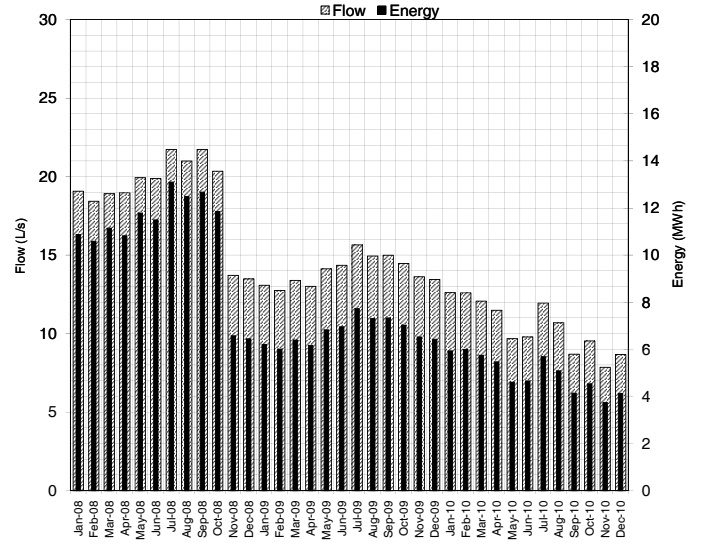


Fig. 4. Average flow rate supplied to the high network (L/s) and energy consumption (MWh) for pumping into the low tank in the period 2008-2010.

in the network. The main advantages of the procedure are, on one hand, the possibility of relying on a calibrated model (of fundamental importance for evaluating beforehand the effects of any operation in the system) and, on the other, the availability of the Pareto optimal solutions representing the different level of compromise between costs and leakage reduction. An application to a real system proved the effectiveness of such approach, as evidenced by the water and energy savings which, together with the reduction of intervention costs due to pipe or joint breaks, allow a more efficient and sustainable management of the system.

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